Mission Control:
From Human Gestures to Intelligent Robot Motion

Creative Inquiry — Fall 2026

Course:	ME 2900-007/3900-107/4900-207
CRN:	92799/92800/92801
Credit hours:	1
Weekly meeting:	Fridays, 10:00 a.m.–12:00 noon, EIB 257
Modality:	In person

Project Advisor(s)

Dr. Umesh Vaidya
Professor
Department of Mechanical Engineering
125A Fluor Daniel Building
uvaidya@clemson.edu
Office hours: By appointment

Sriram Krishnamoorthy, Ph.D.
Postdoctoral Fellow
Department of Mechanical Engineering
121 Fluor Daniel Building
sriramk@clemson.edu
Office hours: By appointment

Required Materials

Students must bring a laptop computer running Windows 11 (Ubuntu 20.04 is preferred). The laptop must have a functioning webcam, reliable Wi-Fi connectivity for robot communication, and permission to install required course software, including Python and Visual Studio Code. All robot hardware, robot-camera access, shared software, and maze materials will be provided. CI course website: website: https://dyco-ai.github.io/go1_gesture_tracking/.

Course Description

This one-credit Creative Inquiry course introduces undergraduate students to human–robot teaming, computer vision, and safe high-level control of a quadruped robot. Students will also gain hands-on experience with the software tools and communication frameworks used for sensing, control, and system integration in modern robotic systems. Students will first develop and test hand-gesture recognition software using the webcams on their personal laptops. During robot operation, the laptop webcam will observe the operator's hand gestures, while a separate live video feed from the camera mounted on the Unitree Go1 quadruped robot will show the operator the environment in front of the robot. Each recognized gesture will be translated into a safe high-level motion command, such as move forward, move backward, turn, or stop. The final objective is to use this remote command interface to guide the Go1 safely through a modular maze without colliding with walls or obstacles.

Students work in teams addressing research and development problems under the supervision of a faculty advisor. Engineering principles and best practices are employed, with emphasis on teamwork, professionalism, experimental practice, documentation, and technical communication. Credits earned in this course may be applied toward the technical elective requirements. Enrollment requires consent of the instructor.

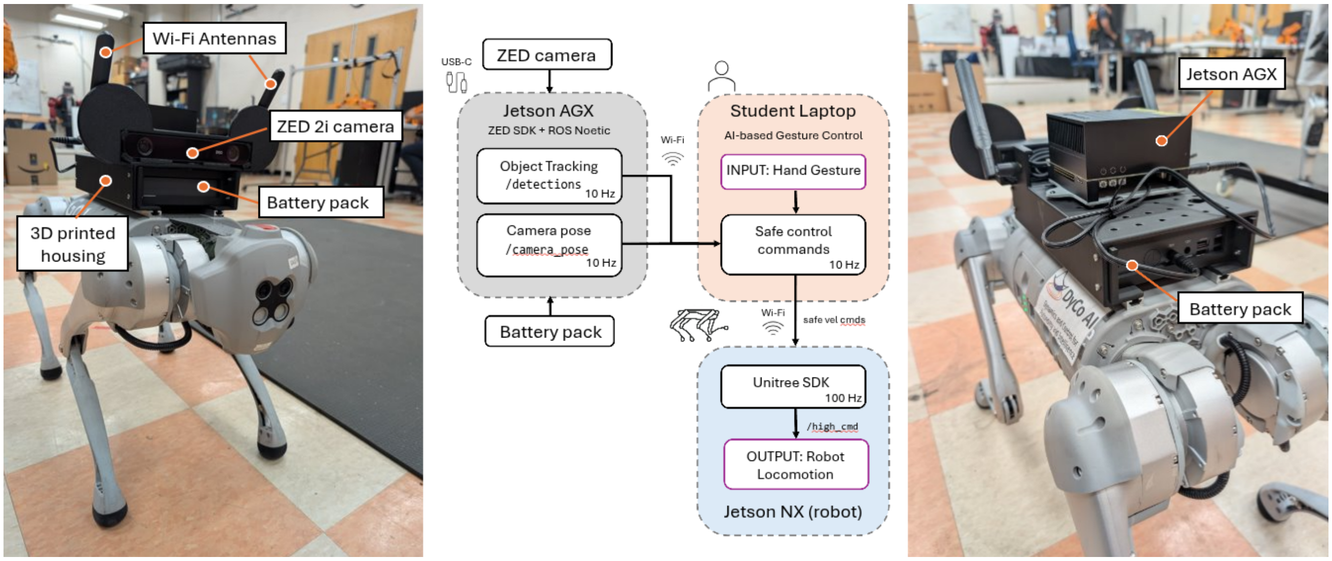

Figure 1: Example robot capabilites: https://sriram-2502.github.io/go1_density_social_nav/

Project Description

Mobile robots are often deployed into cluttered or hazardous environments where a human operator cannot maintain direct visual contact with the robot. Such scenarios arise in applications including search and rescue, disaster response, infrastructure inspection, precision agriculture, and defense operations, where direct human access may be unsafe or impractical. In these situations, the operator must understand the robot's surroundings through onboard video and issue commands from a remote mission-control station. Reliable human–robot collaboration is therefore essential for enabling robots to perform tasks safely and effectively in challenging environments.

This project investigates an AI-enabled hand-gesture interface: the operator views the live video feed from the robot while computer-vision and AI models use the laptop webcam to recognize the operator's gestures. Motion-planning and control-design methods are then used to translate validated gestures into bounded robot velocity commands. A practical remote-command system must also address uncertain gesture recognition, video and communication delays, stale commands, obstacle proximity, and safe robot behavior when information is lost.

Students will develop a gesture-based command interface for the Unitree Go1 quadruped. Using their laptop webcams, teams will implement a starter vocabulary of stop, forward, backward, left, and right commands before integrating the interface with the robot's live video feed for remote maze navigation. Recognized gestures will be converted into bounded high-level velocity commands through the lab's existing control interface, with a safety layer that stops the robot when commands are stale or uncertain, communication or perception is lost, or an obstacle enters a defined stopping region.

Students will work in small teams to extend the starter system. Possible extensions include designing additional gestures, mapping gestures to instructor-approved body poses or gaits, and composing time-limited motion primitives. Each extension must remain interruptible by the stop command and must operate through the approved command and safety interfaces.

The final system will be evaluated in a student-designed testing environment that allows for flexible layouts, obstacles, and task configurations. The operator will remain at a separate command station and navigate using the robot's live video feed rather than direct visual observation of the maze. The required course activity will use a flat maze. Stair climbing and navigation through a debris field may be considered as optional advanced modules only after separate instructor review and validation.

Expected Student Activities

Students will work primarily in teams of two. Students will be expected to:

- Attend and participate in weekly project meetings.
- Learn the necessary introductory Python, computer vision, Git version control, and robot communication concepts required to develop and integrate intelligent robotic systems.
- Develop gesture-recognition software using a laptop webcam.
- Collect or use the provided image and video data to test gesture performance.
- Integrate laptop-based gesture recognition with the live video feed from the robot.
- Map validated gestures to approved, bounded robot actions.
- Implement and test command timeouts, confidence checks, stopping behavior, and other safety logic.
- Design and construct a modular maze within instructor-provided dimensional and safety constraints.
- Participate in integration tests and final maze-navigation trials.
- Maintain an individual research and design journal documenting decisions, results, problems, and contributions.
- Communicate progress through brief weekly updates, a final presentation, and a written report.

Students will begin with laptop-only development and recorded or live camera data. Access to live robot testing will be granted after the required software and safety checks have been completed.

Learning Outcomes

Upon successful completion of the course, students will be able to:

1. Describe the basic architecture of a robotic system, including sensing, computation, communication, control, and actuation.
2. Explain fundamental concepts in human–robot teaming, including how a human operator and a robot share information, decisions, and control responsibilities.
3. Apply basic robot-motion and control concepts, including body-frame velocity, turning rate, gait selection, and high-level command generation.
4. Use and evaluate camera-based AI models for hand detection and gesture recognition.
5. Apply introductory motion-planning and control-design concepts to translate recognized operator intent into bounded, safe robot velocity commands.
6. Integrate robot sensing, human commands, control logic, and robot hardware into a functional mechatronic system.
7. Incorporate safety constraints and fail-safe behavior into robot operation under uncertain perception, communication delays, and nearby obstacles.

8. Design and conduct repeatable engineering tests, analyze recorded data, and use evidence to evaluate performance and justify design decisions.
9. Contribute effectively to a collaborative robotics project using appropriate documentation, version-control, and project-management practices.
10. Communicate the motivation, design, results, limitations, and future directions of a robotics project to technical and nontechnical audiences.
11. Prepare and present project results through a research poster and live robot demonstration for audiences beyond the classroom.

Students will have the opportunity to present their work at Clemson University's annual Focus on Creative Inquiry (FoCI) poster forum. With advisor approval, projects that demonstrate sufficient technical maturity and research contribution may also be developed for submission to appropriate IEEE conferences or robotics competitions.

Weekly Activities Outline

The following 12-week activity outline represents the anticipated project progression. It intentionally leaves flexibility for university holidays, student preparation, make-up work, additional testing, and changes in project progress. The final two activity weeks are reserved for supervised robot demonstrations and maze navigation.

Week	Topic and activity
1	Project introduction, human–robot teaming objective, lab orientation, and robot safety.
2	Development environment, Python fundamentals, version control, and team workflow.
3	Image acquisition and visualization using laptop webcams.
4	Hand detection, landmarks, and the starter gesture vocabulary.
5	Gesture classification, confidence thresholds, temporal consistency, and performance testing.
6	Mapping gestures to abstract robot actions and testing commands in mock mode.
7	Robot camera feed, remote operator display, and high-level robot velocity commands.
8	Command timeouts, default-stop behavior, and obstacle-proximity stopping.
9	Integration of gesture recognition, remote video, motion commands, and safety supervision.
10	Modular maze design, supervised robot integration, and demonstration-readiness review.
11	Robot Demonstration I: system checkout, practice maze navigation, and corrective testing.
12	Robot Demonstration II: final team maze navigation, presentation, and project review.

Course Grading Policy

Students are expected to contribute approximately three hours per week for the one credit hour of enrollment. There are no examinations. Grades will reflect consistent individual engagement, technical progress, teamwork, documentation, safe lab practice, and communication.

Team results will contribute to the course completion grade, but each student must document and explain an identifiable individual contribution. A technically sound, well-documented, and safely tested system may receive full credit regardless of final robot demonstration.

Deliverables

Students will complete the following deliverables:

1. **Weekly progress update:** A concise report/slides of completed work, evidence, challenges, and next steps.
2. **Individual research and design journal:** A chronological record of technical work, decisions, test results, failures, and individual contributions.
3. **Gesture-recognition milestone:** A laptop-webcam demonstration of the starter gesture vocabulary with documented testing.
4. **Mock-command milestone:** Demonstration that gestures produce bounded abstract commands without operating the physical robot.
5. **Safety milestone:** Demonstration of default-stop behavior, command expiration, and obstacle-proximity stopping.
6. **Remote-interface milestone:** Demonstration of laptop gesture recognition operating alongside the live Go1 camera feed.
7. **Team maze design:** An instructor-approved modular maze that satisfies dimensional, visibility, and safety requirements.
8. **Final integration demonstration:** Supervised gesture-based navigation of the Go1 through an approved maze.
9. **Final presentation:** A team presentation describing motivation, design, implementation, testing, results, limitations, and future work.
10. **Final written report/poster:** A technical report or poster that clearly presents the project objectives, methodology, results, and key conclusions.

Lab and Robot Safety

Students must complete all lab and equipment training required by the project advisor before operating the robot or working in the robot test area. The following policies apply:

- Students may not operate any robots without instructor or authorized lab supervision.
- Laptop-webcam, mock-command, and safety milestones must be completed before live robot testing.
- Students may use only instructor-approved high-level robot actions and parameter limits.
- Direct low-level motor commands and arbitrary velocity values are prohibited.
- A designated safety operator must be present during every live robot test with access to the approved emergency-stop method.
- Loose, sharp, unstable, rolling, or otherwise hazardous debris is prohibited.
- Damage, unexpected motion, loss of control, or unsafe conditions must be reported immediately.

Course Absence/Attendance Policy

Attendance and active participation in weekly meetings and scheduled lab sessions are required because the project depends on coordinated team development and supervised access to shared hardware. Students should notify the project advisor as early as possible regarding an anticipated absence and should use the Clemson Notification of Absence process when applicable.

An excused absence does not remove responsibility for communicating with the team, reviewing missed material, and completing an agreed-upon contribution. Two unexcused absences will result in a reduction of one letter grade in the student's final course grade.

AI Policy

Artificial intelligence (AI) tools may be used in this course to support students' understanding of programming concepts, interpret error messages, troubleshoot code, and clarify the operation of algorithms and control methods. AI tools should be treated as learning aids rather than substitutes for independent technical work.

Students must take full ownership of all code, control frameworks, models, analyses, and design decisions submitted for this course. Each student is responsible for verifying the correctness, safety, and performance of any AI-assisted work and must be able to explain, modify, and defend all submitted material. Submitting AI-generated code or technical content without understanding and validating it is not permitted.

Accessibility

Reasonable accommodations will be provided for students with documented disabilities. To receive accommodations, students must register with Student Accessibility Services (SAS) at https://www.clemson.edu/campus-life/campus-services/sds/. All examinations requiring extended time must be completed at the designated testing center. Students are responsible for making the necessary arrangements in advance.

Academic Integrity Statement

As members of the Clemson University community, we uphold Thomas Green Clemson's vision of the University as a "high seminary of learning." Central to this vision is a shared commitment to truthfulness, honor, and responsibility, which are essential for earning the trust and respect of others.

Academic dishonesty diminishes the value of a Clemson University degree. Therefore, lying, cheating, stealing, plagiarism, and other forms of academic misconduct will not be tolerated. Students are expected to follow all University academic integrity policies and guidelines, available at https://www.clemson.edu/academics/integrity/.

All software, reports, figures, datasets, and presentations must properly identify and cite material obtained from external sources. Collaboration within assigned teams is expected, but each student must accurately represent their individual contribution.

Title IX Statement

Clemson University is committed to providing equal opportunity for all individuals and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy, national origin, age, disability, veteran status, genetic information, or protected activity in employment, educational programs and activities, admissions, or financial aid.

This policy includes the prohibition of sexual harassment and sexual violence as required by Title IX of the Education Amendments of 1972. Additional information is available at https://www.clemson.edu/campus-life/campus-services/access/title-ix/.

Ms. Alesia Smith serves as Clemson University's Title IX Coordinator and Executive Director of Equity Compliance. Her office is located at 110 Holtzendorff Hall. She may be contacted by telephone at 864-656-3181 (voice) or 864-656-0899 (TDD).

Emergency Preparedness Statement

Clemson University Emergency Management website and in the purple-covered emergency flipbooks located in offices and classrooms across campus. Students and employees should register to receive CU Alert text messages, become familiar with campus emergency procedures, and follow all instructions provided by emergency responders during an emergency. Additional information about campus safety resources is available at https://www.clemson.edu/CUSafety.